Selendra Tokenomics v3.2

> **Settlement layer for tokenized assets** — Built for Cambodia. Open to all.
>
> 📦 **Source**: github.com/selendra/selendra

Executive Summary

Selendra is a dual-VM blockchain (EVM + WASM) with 1-second finality, designed as the settlement layer for tokenized assets. Starting with loyalty points in Cambodia, the network is open to all builders worldwide.

This document outlines the tokenomics model: strategic scarcity via 320M hard cap, deflationary mechanisms (90% fee burn), and sustainable validator incentives.

What is Selendra?

- **Dual VM Support** — Deploy Solidity (EVM) or ink! (WASM) smart contracts on the same chain
- **1-Second Finality** — AURA + AlephBFT hybrid consensus, no waiting for confirmations
- **Sub-Cent Fees** — Transaction costs under $0.01, affordable for real businesses
- **Unified Accounts** — Single `0x...` address works across all VMs
- **Cambodia First** — KHQR payment integration, local currency settlement

Strategy: Cambodia First, Then ASEAN

```
Phase 1: Cambodia (Current)
├── Loyalty points tokenization (Bitriel)
├── KHQR payment integration
├── CPL fan engagement (StadiumX)
└── Local merchant network

Phase 2: ASEAN Expansion
├── Thailand, Vietnam, Philippines
├── Cross-border settlements
└── Regional payment corridors

Phase 3: Global
├── Emerging markets
├── Enterprise adoption
└── Full decentralization
```

Token Overview

Parameter	Value
Token Name	Selendra

Parameter	Value
Symbol	SEL
Decimals	18
Current Supply	230,000,000 SEL
Burn Amount	80,000,000 SEL (35%)
Post-Burn Supply	150,000,000 SEL
Maximum Cap	320,000,000 SEL (on-chain)
Future Inflation	170,000,000 SEL
Consensus	NPoS (Nominated Proof of Stake)

> **Note**: The 320M cap is already set in the Selendra codebase (`DefaultSelCap` in `primitives/src/lib.rs`). This is an immutable on-chain parameter.

Supply Model

The Burn & Rebuild Strategy

```
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│                                                             │
│   CURRENT         BURN          POST-BURN       MAX CAP     │
│   230M SEL   →   -80M SEL   →   150M SEL   →   320M SEL     │
│                    🔥                                        │
│                                                             │
│   35% burned to create scarcity                             │
│   170M available for future staking rewards                 │
│   ~25-30 year runway to max cap                             │
│                                                             │
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```

The 320M Cap (Already On-Chain)

The 320M maximum supply is hardcoded in the Selendra blockchain:

```
// From github.com/selendra/selendra/primitives/src/lib.rs
pub const DEFAULT_SEL_CAP: u128 = 320_000_000 * TOKEN;
```

Verify on-chain:

```
curl -sX POST https://rpc.selendra.org -H "Content-Type: application/json" \
  -d '{"jsonrpc":"2.0","id":1,"method":"state_getStorage","params":["0x..."]}'
```

At 320M Cap	Value
Price @ $10B MC	$31.25
Staking runway	~25-30 years
vs Avalanche (720M)	2.25x more scarce
vs Cosmos (390M)	1.2x more scarce

Token Distribution

Post-Burn Allocation (150M SEL)

Category	Amount	Percentage	Vesting	Purpose
Angels (10 investors)	15M	10.0%	2yr linear, 6mo cliff	Early supporters
Community (34k holders)	20M	13.3%	Unlocked	BSC community
Team & Founders (5 people)	15M	10.0%	4yr linear, 1yr cliff	Core team retention
DAO Treasury	50M	33.3%	Governance-controlled	All development & grants
Staking Rewards Pool	20M	13.3%	Emission schedule	Initial validator rewards
DEX Liquidity	10M	6.7%	LP locked 2 years	Initial DEX liquidity
Ecosystem Grants	20M	13.3%	DAO proposals	Developer incentives
Total	**150M**	**100%**		

> **No Foundation**: All treasury funds are controlled by on-chain governance (Democracy + Treasury pallets). No centralized entity holds tokens.

Visual Distribution

```
┌──────────────────────────────────────────────────────────────┐
│                                                              │
│  Angels           ████████████ 10.0%                         │
│  Community        █████████████████ 13.3%                    │
│  Team             ████████████ 10.0%                         │
│  DAO Treasury     ██████████████████████████████████ 33.3%   │
│  Staking Pool     ████████████████ 13.3%                     │
│  DEX Liquidity    ████████ 6.7%                              │
│  Ecosystem        ████████████████ 13.3%                     │
│                                                              │
│  DAO-Controlled: 60.0% ✅                                     │
│  (DAO Treasury + Ecosystem + Staking)                        │
│                                                              │
```

```
│  Community Total: 73.3% ✓                                  │
│  (Community + DAO + Ecosystem + Staking + DEX)             │
│                                                            │
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```

Inflation Schedule

How Selendra Inflation Works (Exponential Decay)

Selendra uses an **exponential decay** model — the same mathematical principle that governs natural processes like radioactive decay. In simple terms:

> **"The closer you get to the cap, the slower the emissions."**

This is implemented on-chain in `ExponentialEraPayout` and controlled by two parameters:

Parameter	Value	Meaning
`SelCap`	320,000,000 SEL	Maximum supply (hard cap)
`ExponentialInflationHorizon`	~4.9 years	Decay speed constant (see below)

Understanding the Horizon (~4.9 years)

The horizon is **not** a countdown or a cycle — it's a **mathematical constant** that controls how fast inflation decays.

In exponential decay, the horizon is the time it takes to fill **~63.2%** of the remaining gap between current supply and the cap. This then repeats for the remaining gap:

```
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│                                                            │
│   After 1 horizon (~4.9 years):  63.2% of gap filled       │
│   After 2 horizons (~9.8 years): 86.5% of gap filled       │
│   After 3 horizons (~14.7 years): 95.0% of gap filled      │
│   After 5 horizons (~24.5 years): 99.3% of gap filled      │
│                                                            │
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Concrete Example (Starting at 150M SEL):

Time	Gap Remaining	Supply	What Happened
Start	170M (320M - 150M)	150M	Initial state
+4.9 years	~63M	~257M	Filled 63% of 170M gap
+9.8 years	~23M	~297M	Filled 63% of remaining 63M
+14.7 years	~8.5M	~311M	Filled 63% of remaining 23M

Time	Gap Remaining	Supply	What Happened
+24.5 years	~1.2M	~319M	Nearly at cap

> The horizon value (`154,283,512,497 ms ≈ 4.89 years`) produces ~5% inflation in year 1, decreasing smoothly over time.

The Formula

```
New Tokens per Era = (1 - e^(-time/horizon)) × (SelCap - CurrentSupply)
```

Benefits: Cap guaranteed (mathematically impossible to exceed 320M), self-adjusting, smooth curve, governance-tunable via `aleph.set_inflation_parameters()`. Same model used by Aleph Zero.

Approximate Year-by-Year Projection

These are approximations — actual rates depend on staking participation and era timing.

Year	~Supply	~Effective Rate	~New Tokens	Notes
1	150M → 158M	~5.0%	~7.5M	Early growth phase
2	158M → 165M	~4.4%	~7.0M	Rate auto-decreases
3	165M → 172M	~3.9%	~6.4M	As gap shrinks
5	177M → 187M	~3.0%	~5.5M	Smooth decline
10	206M → 215M	~2.0%	~4.4M	Approaching maturity
15	228M → 235M	~1.5%	~3.5M	Long-term stability
20	247M → 252M	~1.0%	~2.5M	Near terminal
30	273M → 276M	~0.5%	~1.5M	Asymptotic approach
50+	310M → 320M	~0.1%	<0.5M	Effectively capped

Inflation Curve (Exponential Decay)

```
Inflation %
    │
 5% ┤ ●
    │  \
 4% ┤   \
    │    \
 3% ┤     \
    │      \\
 2% ┤        \\
    │          \\\
 1% ┤             \\\\
```

```
     │                \\\\\\
0.5%┤                      \\\\\\\\→ asymptotic to 0%
     │
     └──────────────────────────────────────────────────────→
       1   5    10    15    20    25    30    40    50+ years

    SMOOTH CURVE: No sudden drops, mathematically guaranteed cap
```

Validator vs Treasury Split

From each era's inflation rewards:

```
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│                                                          │
│   Era Payout Split (on-chain constant):                  │
│                                                          │
│   ├── 75% → Validators & Nominators (VALIDATOR_REWARD)   │
│   └── 25% → Treasury (for ecosystem growth)              │
│                                                          │
│   Note: This split is adjustable via runtime upgrade     │
│                                                          │
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```

Adjusting Inflation via Governance

The inflation model can be tuned without a runtime upgrade:

```
// Governance call to adjust inflation parameters
aleph.set_inflation_parameters(
    sel_cap: Option<Balance>,           // Change the cap
    horizon_millisecs: Option<u64>      // Change decay speed
)
```

Adjustment	Effect
↑ Increase horizon	Slower decay, more inflation
↓ Decrease horizon	Faster decay, less inflation
↓ Decrease cap	Less total supply possible
↑ Increase cap	Not recommended (requires governance)

Post-Cap Economics

What Happens When Supply Reaches 320M?

When `total_issuance >= sel_cap`, the exponential decay formula produces **zero new tokens**. The network transitions from inflation-funded to fee-funded security.

```
┌──────────────────────────────────────────────────────────────┐
│                                                              │
│   BEFORE CAP                │         AT/AFTER CAP           │
│   ──────────────────────────┼───────────────────────────     │
│   Validators earn:          │   Validators earn:             │
│   • Inflation rewards       │   • Transaction fees only      │
│   • Transaction fees        │   • No new token minting       │
│                             │                                │
│   Supply: Growing           │   Supply: Stable or Deflationary │
│                                                              │
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```

At cap, validator incentives come entirely from transaction fees (see Deflationary Mechanisms for fee split details). Burns can push supply **below** 320M.

Self-Balancing Supply

Post-cap, the supply oscillates based on network activity:

```
          High Activity              Low Activity
          ─────────────              ────────────
          Burns > 0                  Burns ≈ 0
          Supply ↓ (deflation)       Supply = 320M (stable)

   Supply
   320M ────────────●─────────────────────────────────────
                    \      /\      /\
                     \    /  \    /  \       ← Oscillates based
                      \  /    \  /    \        on network usage
                       \/      \/      \
   310M ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─
        │
        └────────────────────────────────────────────────→
          Cap reached                            time
```

Comparison to Other Networks

Network	Post-Cap Model	Validator Incentive
Bitcoin	Fixed 21M cap, fees only	Transaction fees
Ethereum	No cap, but burns offset	Inflation + fees - burns
Cosmos	Perpetual 7-20% inflation	Inflation rewards
Selendra	320M cap + fee burns	Fees (deflationary possible) ✅

Timeline Estimate

Phase	~Year	Supply	Validator Revenue Source
Growth	1-15	150M → 280M	~80% inflation, ~20% fees
Transition	15-25	280M → 310M	~50% inflation, ~50% fees
Maturity	25-35	310M → 320M	~10% inflation, ~90% fees
Post-Cap	35+	≤320M	100% fees (deflationary)

Deflationary Mechanisms

1. Initial Burn (One-time)

Parameter	Value
Amount	80,000,000 SEL (35% of current)
Source	Team (40M) + Angels (40M) pre-distribution
Method	Sent to `0x000...dead` in genesis block
Verification	On-chain, provable via block explorer

2. Transaction Fee Burns (Ongoing)

```
┌─────────────────────────────────────────────────────────────┐
│                                                             │
│  EVERY TRANSACTION ON SELENDRA                              │
│                                                             │
│  Gas Fee Split:                                             │
│  ├── 90% → BURNED 🔥 (permanent removal)                    │
│  ├── 7%  → Block Validators                                 │
│  └── 3%  → DAO Treasury                                     │
│                                                             │
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```

3. DEX & Bridge Burns

```
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│                                                             │
│  DEX SWAP FEE (0.30%):                                      │
│  ├── 0.20% → LP Providers (incentive)                       │
│  ├── 0.05% → BURNED 🔥                                      │
│  └── 0.05% → DAO Treasury                                   │
│                                                             │
│  BRIDGE FEE (0.50%):                                        │
│  ├── 0.25% → BURNED 🔥                                      │
```

```
│   └── 0.25% → DAO Treasury                                        │
│                                                                   │
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```

Projected Burn Impact

Scenario	Annual Tx Volume	Fee Revenue	Annual Burn	Net Inflation
Low activity	$10M	$30K	~1M SEL	4.3% (Year 1)
Medium activity	$100M	$300K	~10M SEL	0% (breakeven)
High activity	$1B	$3M	~100M SEL	**-2% deflationary**

At scale, Selendra becomes deflationary like Ethereum post-merge.

Staking Economics

Validator Requirements

Parameter	Current Value	Target Value
Minimum stake (Validator)	25,000 SEL	1,000,000 SEL
Minimum stake (Nominator)	100 SEL	100 SEL
Unbonding period	14 eras (~14 days)	28 eras (~28 days)
Max validators	1,000 (expandable)	1,000 (expandable)
Sessions per era	96 (15 min each)	96 (15 min each)
Nominators per validator	1 (DPoS mode)	1 (DPoS mode)

> **Note**: "Current Value" reflects what's in the codebase today. "Target Value" is the Tokenomics v3.0 goal requiring a runtime upgrade.

Slashing (Penalties)

Offense	Penalty	Notes
Equivocation (double-signing)	10% of stake	Immediate slash
Unresponsiveness	0.1% per era offline	Cumulative
Invalid blocks	Up to 100%	Severe, rare

> Slashed funds go to Treasury. Nominators share the slash proportionally.

Reward Distribution

From each era's inflation: **75%** → Validators & Nominators | **25%** → Treasury (defined as `VALIDATOR_REWARD` in runtime).

APY Projections

Staking Rate	Validator APY	Nominator APY	Interpretation
30%	18%	15%	Need more stakers
50%	12%	10%	Below target
65%	**10%**	**8%**	**Target ✅**
80%	7%	5%	Above target
90%	5%	3%	Over-staked

Dynamic adjustment: APY increases when staking is low, decreases when high.

Vesting & Anti-Dump

Allocation	Cliff	Vesting	Full Unlock
Angels (15M)	6 months	Linear	24 months
Team (15M)	12 months	Linear	48 months
DAO Treasury (50M)	None	Governance-controlled	On-demand

Bridge Friction (BSC → Selendra): 0.5% fee, 10K SEL minimum, 15min delay, 1M daily limit per wallet.

Governance

All treasury funds controlled by on-chain governance. **No foundation.**

Active Pallets: democracy (32), treasury (16), collective (30/31), vesting (53), preimage (34)

Needs Adding: `pallet_bounties`, `pallet_tips` (requires runtime upgrade)

Proposal Type	Quorum	Approval	Voting Period
Parameter change	5%	50%+1	7 days
Treasury spend (<1M)	10%	50%+1	14 days
Treasury spend (>1M)	20%	66%	28 days
Protocol upgrade	25%	75%	28 days

Voting: 1 SEL = 1 Vote. Standard Substrate conviction voting (0.1x–6x multiplier based on lock period).

Technical Comparison

Metric	Selendra	Ethereum	Solana	Avalanche

Metric	Selendra	Ethereum	Solana	Avalanche
VM Support	EVM + WASM	EVM	SVM	EVM
Block Time	1 second	12 seconds	0.4 seconds	2 seconds
Finality	1 second	15 minutes	12 seconds	2 seconds
Max Supply	320M	Uncapped	Uncapped	720M
Current Supply	150M*	120M	570M	450M
Fee Burn	90%	~80%	50%	100%
Consensus	NPoS + AlephBFT	PoS	PoH + PoS	Avalanche

*Post-burn

Risks & Mitigations

Risk	Impact	Mitigation
Low fee revenue post-cap	Validators leave	Governance can reduce burn rate (90% → 70%) or add micro-inflation
Validator centralization	1M SEL requirement too high	Target value adjustable via governance; start lower
BSC dump at migration	Price crash	Bridge friction (0.5% fee, 15min delay, daily limits)
Governance attack	Treasury drained	20-25% quorum + 66-75% approval for large spends
Smart contract exploit	Loss of funds	Audits required before mainnet; bug bounty program

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Document Info

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